

Project Initialization and Planning Phase

Date	12 July 2026
Team ID	6a2fbf036b1ed0bd8fa789aa
Project Title	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to simplify and automate debt management for borrowers using AI-driven financial analysis, boosting settlement success rates and easing negotiation stress. It tackles the lack of accessible, personalized debt-relief tools, promising better financial outcomes, reduced default risk, and more confident borrowers. Key features include an AI-powered settlement engine and Gemini-driven negotiation letter generation.

Project Overview	
Objective	The primary objective is to simplify and automate debt management for borrowers by providing AI-powered financial health analysis, settlement prediction, and negotiation support through a secure digital platform.
Scope	The project covers borrower onboarding, loan and financial data management, AI-driven settlement recommendations, Gemini-powered negotiation letter generation, and financial health tracking within a full-stack web application.
Problem Statement	
Description	Borrowers struggling with overdue loans often lack the financial literacy, tools, and negotiation skills needed to secure fair settlements with lenders, leading to prolonged debt stress and risk of default.
Impact	Solving this will empower borrowers to make informed financial decisions, improve their chances of successful debt settlement, and reduce default rates — benefiting both borrowers and lenders.
Proposed Solution	
Approach	Employing AI (Google Gemini) alongside rule-based financial analysis to assess borrower profiles, predict feasible settlement amounts, and generate personalized negotiation strategies and letters.
Key Features	- Implementation of an AI-powered settlement prediction and debt stress analysis engine. - Gemini AI-integrated negotiation letter generator for lender-specific communication. - Real-time financial health dashboard tracking EMI ratio, monthly surplus, and stress level. - Secure JWT-based authentication and negotiation history tracking, with rule-based fallback when the AI service is unavailable.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	2 vCPUs (standard cloud instance; no GPU required)
Memory	RAM specifications	4 GB
Storage	Disk space for data, logs, and application files	20 GB SSD
Software		
Frameworks	Backend and frontend frameworks	FastAPI (Python), React.js + Vite
Libraries	Additional libraries	SQLAlchemy, Pydantic, python-jose (JWT), google-generativeai (Gemini SDK), axios, recharts
Development Environment	IDE / tools	VS Code, Postman (API testing)
Data		
Data	Source, size, format	Borrower-entered loan & financial data (user input); synthetic/sample loan datasets for testing, csv format